

Chapter 1 — Nutrition in Plants

1. What is Nutrition?

Nutrition is the process by which organisms obtain food and use it for energy, growth, repair and maintenance.

Plants need nutrients such as water, minerals and carbon dioxide.

2. Modes of Nutrition

There are two broad modes:

Autotrophic nutrition

Green plants prepare their own food using carbon dioxide and water in the presence of sunlight and chlorophyll.

Heterotrophic nutrition

Organisms that cannot make their own food obtain it from other organisms or organic matter.

3. Photosynthesis

Photosynthesis is the process by which green plants prepare food using **carbon dioxide and water** in the presence of **sunlight and chlorophyll**.

A simplified word equation is:

Carbon dioxide + Water → Glucose + Oxygen

Conditions required:

-  Sunlight
-  Chlorophyll
-  Carbon dioxide
-  Water

The food produced is commonly represented as **glucose**. Some of it can be stored as **starch**.

4. Role of the Leaf

Leaves are the main organs where photosynthesis takes place in most green plants.

They contain **chlorophyll**, the green pigment that absorbs light energy.

5. Stomata

Stomata are tiny pores found mainly on the surfaces of leaves.

They allow gases to move into and out of the leaf.

- Carbon dioxide enters for photosynthesis.
- Oxygen produced during photosynthesis can leave.
- Water vapour can also escape through them.

Guard cells control the opening and closing of stomata.

6. Other Modes of Nutrition

Type	Description	Example
Parasitic	Obtains food from a living host	Cuscuta
Saprotrophic	Obtains nutrients from dead and decaying matter	Mushroom
Insectivorous	Traps insects to supplement nutrients	Pitcher plant
Symbiotic	Two organisms live together and benefit from the association	Lichen

7. Important Examples

Cuscuta:

A parasitic plant that obtains food from a host plant.

Pitcher plant:

An insectivorous plant that traps and digests insects. This helps it obtain nutrients such as nitrogen.

Lichen:

A symbiotic association between an alga and a fungus. The alga makes food, while the fungus provides support and helps with water and minerals.

★ Quick Revision

Remember:

Photosynthesis =

☀ Sunlight

- 🌿 Chlorophyll
 - 🌬 Carbon dioxide
 - 💧 Water
- 🍷 Food + 🌬 Oxygen

Key terms

- Nutrition

- Autotroph
 - Heterotroph
 - Photosynthesis
 - Chlorophyll
 - Stomata
 - Parasite
 - Saprotroph
 - Insectivorous plant
 - Symbiosis
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Test Yourself

1. Why are green plants called autotrophs?
2. What is photosynthesis?
3. What is the role of chlorophyll?
4. What are stomata?
5. Why is Cuscuta called a parasite?
6. Why do insectivorous plants trap insects?
7. What is saprotrophic nutrition?
8. What is symbiosis?

EduLearn Challenge

A plant is kept in darkness for several days. What would happen to its ability to make food, and why?